

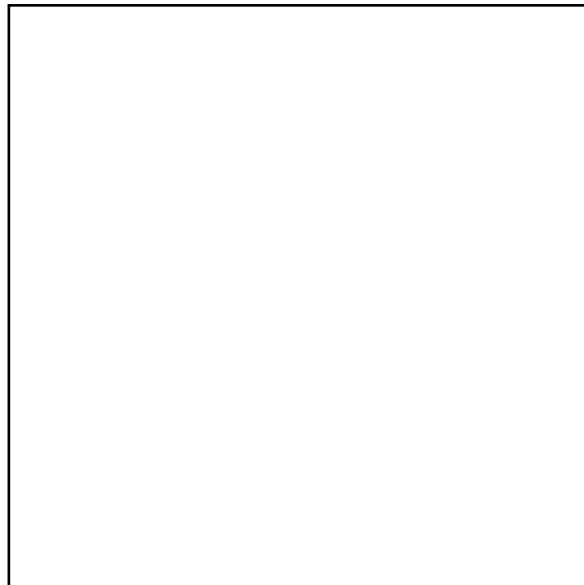
Name:

Student number:



EEE3094S 2023: Class Test 2

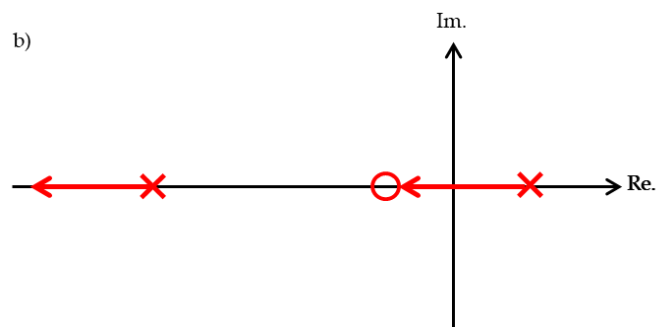
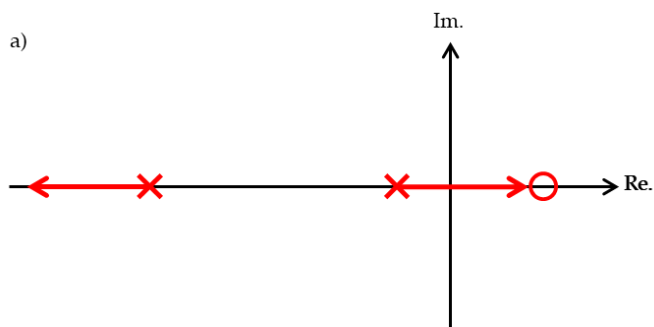
- The test is open book
- Answer on the test sheet only. You may use the provided with extra working space on the back side of each page, but please note that if you have done so (e.g. write “see opposite page” by the question) else your answers may be missed while marking.
- There are four questions totalling 50 marks.
- You have 90 minutes to complete the test.
- Answer space is not indicative of the length of answer expected, you have intentionally been given excessive space.
- The last page is an information sheet.
- The information sheet includes a photo of my cat, Mouse. You are invited to draw a picture of her in the space below if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make me and the tutor marking your test happy, which might make a difference in your favour. **Also, the “best” Mouse drawing will gain the artist an unspecified reward.**



[illegible]

Question 1 [6]

The Nyquist plot of an unknown system encircles -1 once in a clockwise direction. What does this tell you about the stability of the closed-loop system, if the root locus of the system is as shown in each of the following cases? [2 x 3 marks]

[illegible]

[illegible]

Question 2 [22]

The transfer function relating the position to the field voltage of a field-controlled DC motor is

$$G(s) = \frac{A}{s(1 + s\tau_m)(1 + s\tau_e)}$$

where A is the gain of the motor, τ_m is a mechanical time constant related to the mechanical damping and inertia of the motor shaft, and τ_e is the electrical time constant related to the resistance and inductance of the field coil.

Because the mechanical time constant is often much larger than the electrical one, the reduced-order model $G_r(s) = \frac{A}{s(1+s\tau_m)}$ is often a reasonable simplification.

You may use $A = 1$, $\tau_m = 0.5$ and $\tau_e = 0.05$.

- a) Does the reduced-order model behave the same way as the original model with respect to closed-loop stability? You may support your answer with rough sketches of a suitable plot. [4]

Plot space:

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[illegible]

You decide to place this motor in feedback with a proportional controller to have it track a position input.

- b) Evaluate whether it is possible to meet the following design specifications. Give brief explanations supporting your decisions:
- i) Tracking of a position with >90% accuracy. [2]
 - ii) Perfect correction of a constant offset in the output position. [2]
 - iii) Settling time less than the mechanical time constant. [2]

[illegible]

- c) A design specification states that the response of the system should have no more than 10% overshoot. Using the reduced-order model, identify a position for the closed-loop poles that gives the fastest possible settling time and 10% overshoot. Find the value of K needed to place the closed-loop poles there. Would you recommend using this gain value? [8]

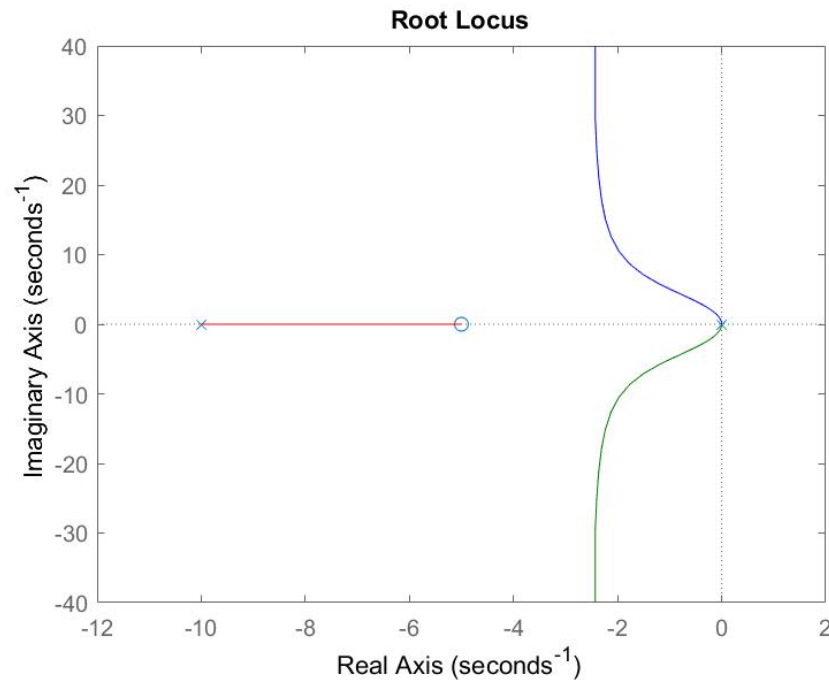
[illegible]

[illegible]

[illegible]

Question 3 [10]

Consider the root locus of a system shown below:



Based on this plot, decide whether the given statements about the system are true or false. Support your answers with brief explanations.

- a) The system has infinite gain margin when placed in feedback. [2]

- b) The system will not be affected by input or output disturbances that can be modelled as a constant velocity offset. [3]

- c) The system will behave more like a second-order system with increasing gain. [2]

- d) Adding another zero between -4 and the origin will cause the system to behave like an overdamped second-order system at high gains. [3]

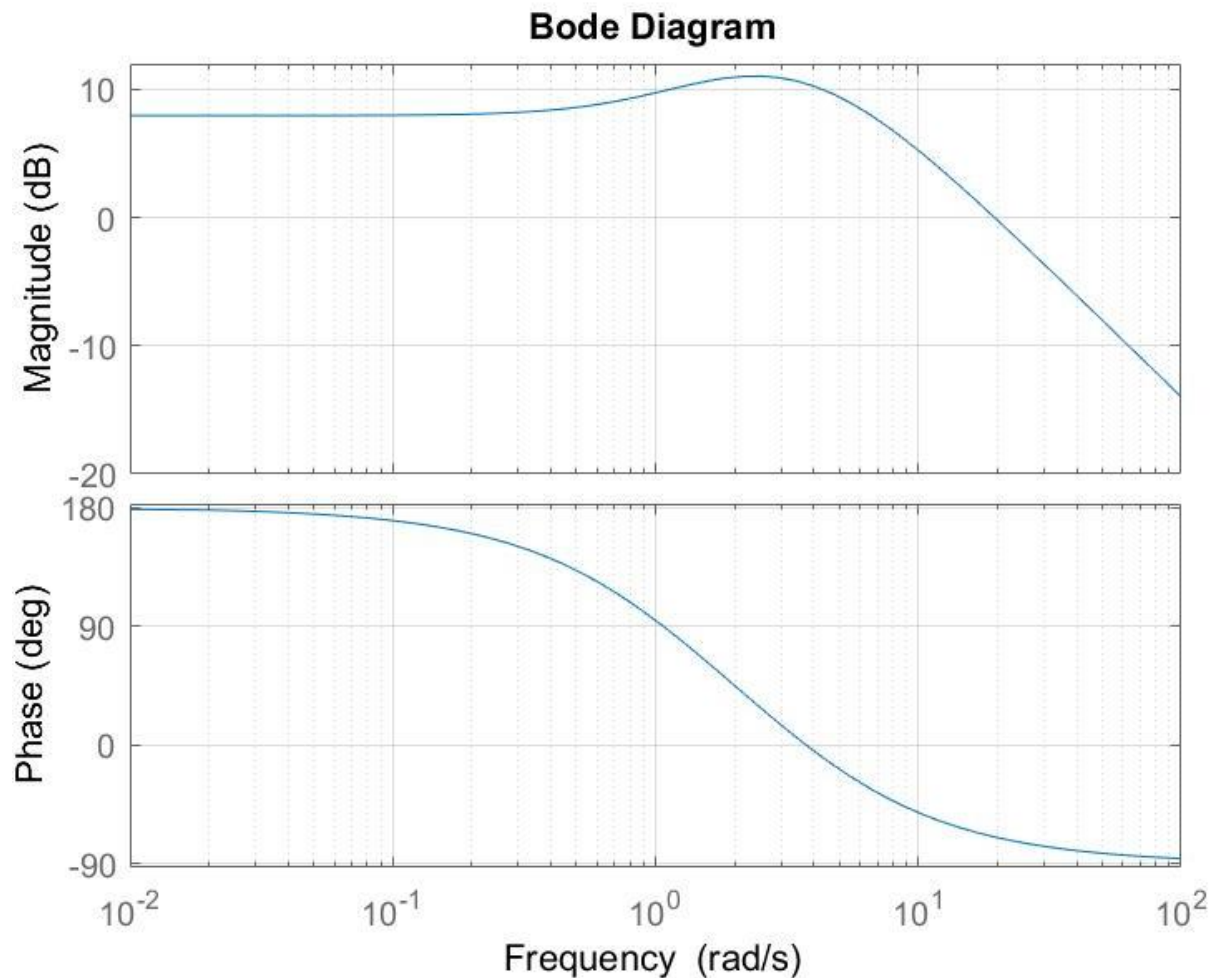
[illegible]

Question 4 [12]

The given Bode plots show the frequency responses of systems that are stable in open loop.

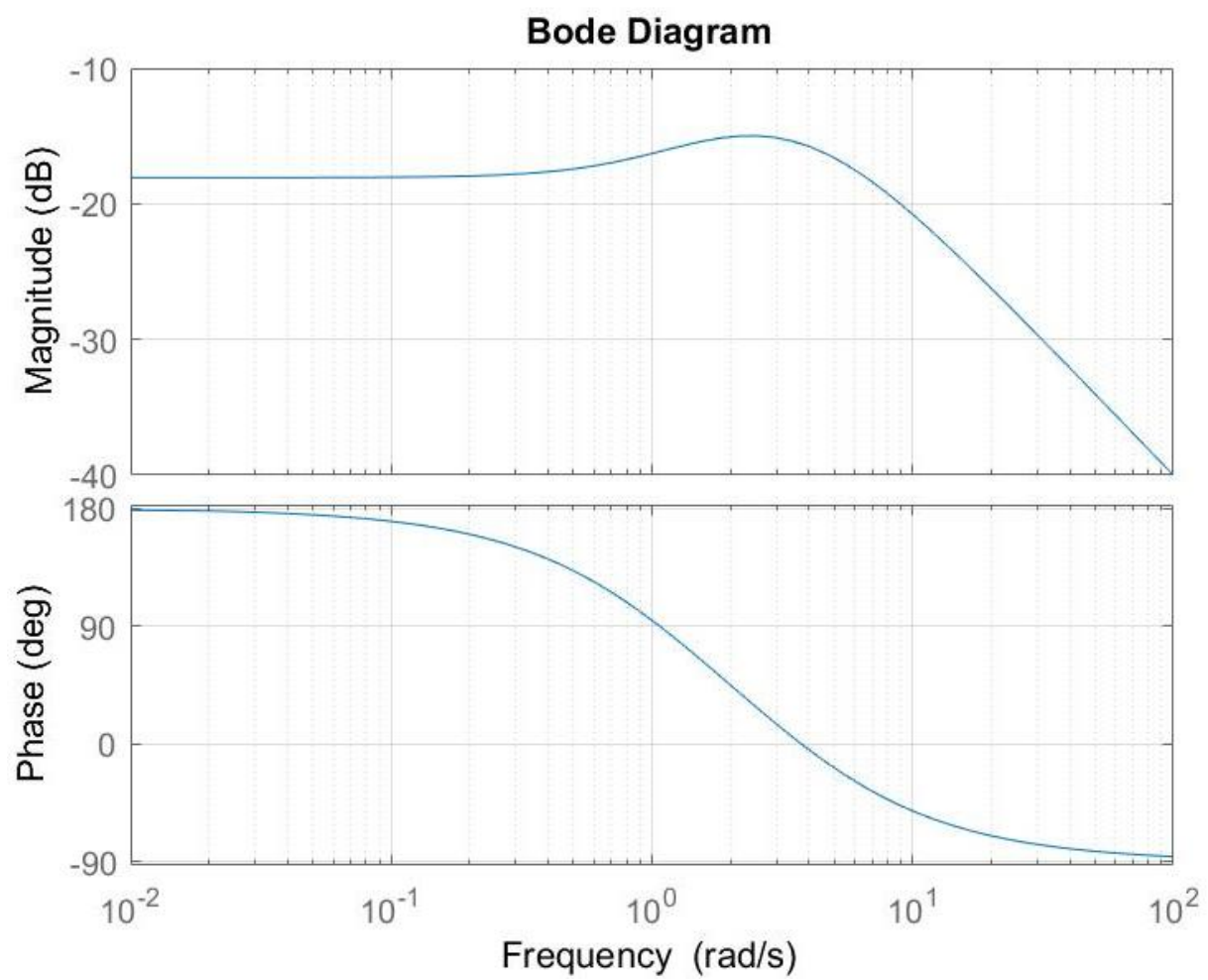
For each of these plots, determine the gain and phase margin of the system, and annotate the plot to illustrate your working (i.e. indicate the gain and phase crossovers where applicable). Evaluate whether the closed-loop system is stable, and if so, whether a proportional controller could cause it to become unstable. [3 x 4 marks]

a)

[illegible]

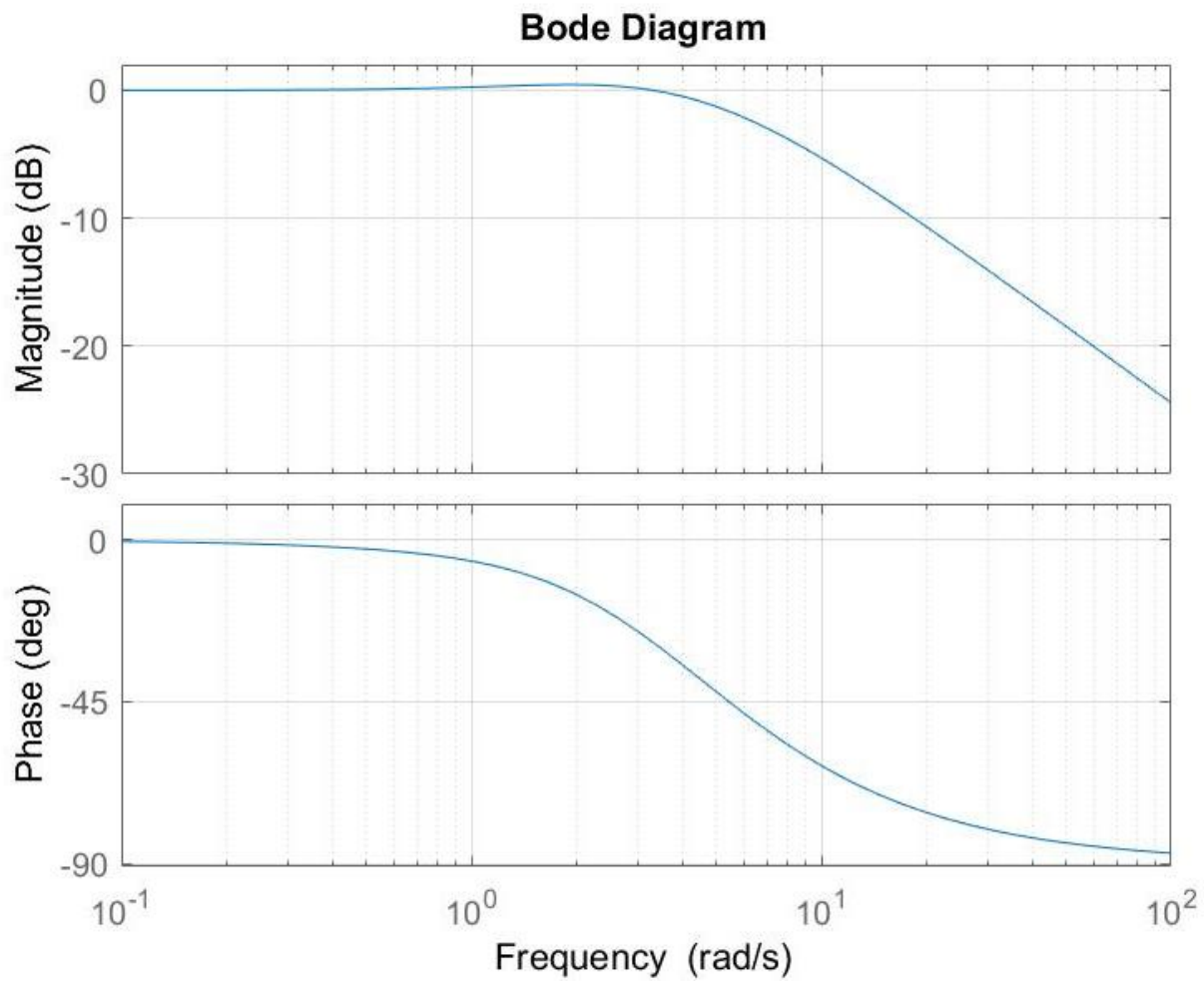
[illegible]

b)

[illegible]

[illegible]

c)

[illegible]

[illegible]

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} \right)}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>

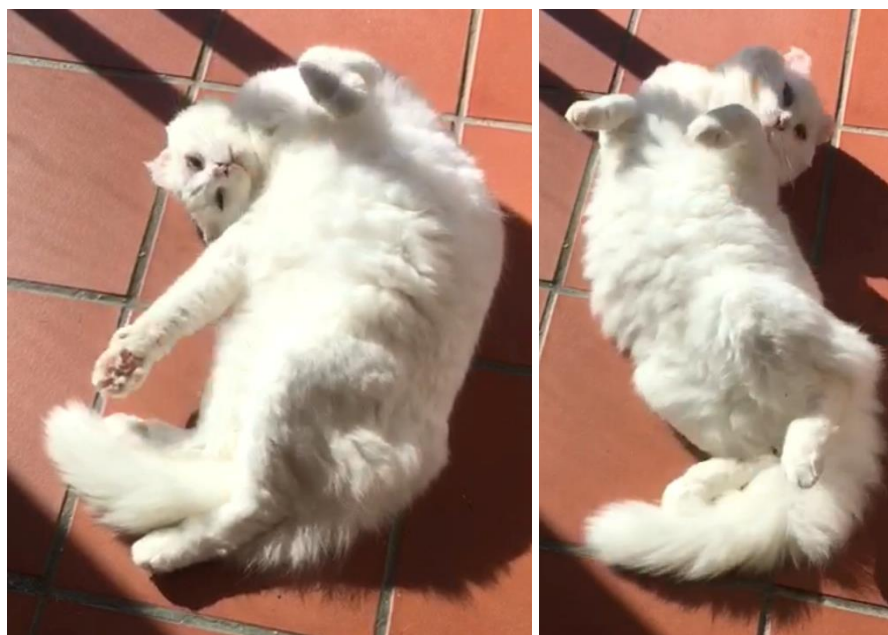


Fig. 1 Mouse on a roll